Here are the answers to all the questions on the worksheet:

Page 1: Group I Elements

(a) (i) Complete the table to estimate:

- **Relative thermal conductivity of potassium:** Any value between 1.7 **and** 3.8 (e.g., 2.5). (Reasoning: conductivity decreases down the group from sodium to caesium).
-
- **Atomic radius of rubidium:** Any value between $228pm$ **and** $264pm$ (e.g., $248pm$). (Reasoning: atomic radius increases down the group).
-

(a) (ii) Describe the trend in the boiling points of the Group I elements.

- The boiling points decrease down the group.
-

(a) (iii) Use the information in the table to predict what you would observe when rubidium reacts with cold water.

- Extremely rapid bubbling, catches fire immediately with a violent explosion / violent explosion.
-

Page 2: Reactivity Series Investigation

(a) Name the substance that causes the brown coating on the zinc.

- **Copper** (or Cu).
-

(b) State why there is no change in the experiment with zinc nitrate solution.

- A metal cannot displace itself / zinc does not react with zinc nitrate solution.
-

(c) Place the metals aluminium, copper, zinc, and M in order of decreasing reactivity:

1. **Aluminium** *(most reactive)*
2.
3. **Zinc**
4.

5. **M**
6.
7. **Copper** *(least reactive)*
8.

Page 3: Reactivity with Oxygen & Atomic Structure

(d) List these metals in order of their reactivity (least reactive first):

1. **Tungsten** *(least reactive)*
2.
3. **Cobalt**
4.
5. **Iron**
6.
7. **Magnesium** *(most reactive)*
8.

Question 4: Complete Table 2.1 for the Na^+ ion

- **Number of protons:** 11
-
- **Number of neutrons:** 12 $(23 - 11 = 12)$
-
- **Number of electrons:** 10 $(11 - 1 = 10)$
-

Question 4 (continued): Dot-and-cross diagram for methane (CH_4)

- Draw **one single pair of shared electrons** (one dot from Carbon and one cross from Hydrogen) in each of the four overlapping region circles between the central Carbon atom (C) and the four outer Hydrogen atoms (H).
-

Pages 4–5: Chemical Bonding & Ionic Structures

Question 5: Explain why sodium chloride ($NaCl$) and zinc chloride ($ZnCl_2$) contain different numbers of chloride ions.

- Sodium forms a Na^{+} ion (charge $+\ 1$), which requires one Cl^{-} ion (charge $-\ 1$) to balance charges.
-
- Zinc forms a Zn^{2+} ion (charge $+\ 2$), which requires two Cl^{-} ions (charge $-\ 1$) to balance the charge and make a neutral compound.
-

Question 6: Which statement about the boiling points of covalent compounds such as CO_2 and ionic compounds is correct?

- **D** — Ionic compounds have higher boiling points than these covalent compounds because the attractive forces between ions are stronger than the attractive forces between covalent molecules.
-

Question 7: Describe the lattice structure of sodium chloride.

- A **regular repeating three-dimensional giant ionic lattice** consisting of alternating positive sodium ions (Na^{+}) and negative chloride ions (Cl^{-}) held together by strong electrostatic forces of attraction.
-

Question 8: Which row describes the formation of magnesium chloride and the strength of the attraction between its ions?

- **C** — Formation: Electrons are transferred from magnesium to chlorine; Strength of attraction: Strong.